WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – III : IRRIGATION & WATER REQUIREMENT OF CROPS

I. Irrigation – Introduction

> Irrigation = the artificial application of water to soil to supply moisture essential for plant growth when rainfall is insufficient or uneven.

- India: net sown area ≈ 140 M ha; net irrigated ≈ 70 M ha (~ 50%); irrigation potential created ≈ 140 M ha; major sources – canals (≈ 25%), tube wells & wells (≈ 60%), tanks.
- Irrigation engineering = planning, design & operation of works (dams, weirs, canals, regulators, drains).

❖Necessity & importance

- Insufficient rainfall (arid/ semi-arid: Rajasthan, Rayalaseema); uneven distribution in time (monsoon only 3–4 months) and place; growing perennial crops (sugar cane, banana) needing water year-round.
- Commercial crops (cotton, rice, sugarcane) need assured water; improves crop yield (irrigated yield 2–3× rain-fed); allows multiple cropping.
- Food security, green revolution, drought protection (famine), rural employment & income, hydropower & navigation from multipurpose projects, recharge of ground water.

❖Advantages of irrigation

- Increase in crop yield & food production; cash crops; protection from famine; increased land value, general prosperity; hydropower generation, drinking water & navigation; ground water recharge; employment; improves climate; afforestation; revenue to government.

❖Ill-effects of irrigation

- Water-logging (rise of water table) & salinity/ alkalinity (soil degradation); climate change locally (humidity ↑), mosquitoes → malaria & water-borne diseases; over-irrigation → seepage losses; dampness in houses; cost of drainage; rise of water table due to canal seepage; submergence & displacement by dams; unequal distribution of water (head vs tail reaches).

2. Types of Irrigation

- Based on source/ scheme:
  - Surface irrigation – gravity flow using canals (perennial from rivers with storage; inundation canals during floods), tanks (A.P. – small tanks), lift irrigation (pumps from rivers/ canals).
  - Ground-water irrigation – open wells, tube wells, bore wells.
  - Rain-water harvesting/ micro-irrigation.
- Based on size (CCA): Major (> 10,000 ha), Medium (2,000–10,000 ha), Minor (< 2,000 ha) – e.g., Nagarjuna Sagar, Srisailam (major), tanks (minor).
- Based on nature of supply: Perennial irrigation (canal water throughout the year – storage reservoir/ weir with assured flow) and Inundation/ flood irrigation (river flood water drawn without head works) & non-perennial.
- Based on purpose: productive (revenue), protective (famine protection – seasonal).
- Based on rainfall dependence: supplemental irrigation, full irrigation, pre-sowing (paleo).
- Lift vs flow irrigation.

3. Methods of Application of Irrigation Water

> Water is applied to fields by (A) surface irrigation, (B) sub-surface irrigation, (C) sprinkler irrigation, (D) drip (trickle) irrigation.

❖A. Surface irrigation (flooding)

- Free flooding (wild flooding): water let on field freely, uncontrolled; for cheap, undulating land; wasteful.
- Border strip method: field divided into long strips (border strips) 3–15 m wide, 100–400 m long, separated by ridges; water enters at upper end, flows as a sheet; for wheat, fodder, close-growing crops; slope 0.2–1%.
- Check basin method: field divided into small level basins with bunds; water ponded to required depth; best for flat, permeable soils; paddy; orchards; high efficiency.
- Furrow irrigation: small channels (furrows) between crop rows (30–90 cm spacing); water flows along furrows and seeps laterally into ridges; for row crops – maize, sugarcane, cotton, potato, vegetables; less evaporation, wetting limited. Types: straight, contour, corrugation.

- Basin irrigation (ring basin) for orchards (individual trees).

❖B. Sub-surface irrigation

- Water applied below the ground to raise the water table (natural sub-irrigation) or through buried perforated pipes/ tile drains – used in sandy soils with high water table; low evaporation.

❖C. Sprinkler irrigation

- Water pumped under pressure (2–4 kg/cm^2) through pipes & rotating nozzles, sprayed like artificial rain.
- Suitable for: undulating land, sandy/ highly permeable soils, shallow soils, water-scarce areas, frost protection.
- Advantages: saves 30–50% water, no land levelling, uniform, fertigation, efficiency 70–80%. Disadvantages: high initial cost (pumps, pipes), wind affects uniformity, energy cost, not suited to heavy clay, evaporation in hot wind.

❖D. Drip (trickle) irrigation

- Water delivered drop by drop directly to the root zone through emitters (drippers) on laterals fed by mainline, sub-main, filters, fertiliser tank, pump; discharge 2–8 L/h.
- Efficiency 90–95%; saves 40–70% water; reduces weeds, fertigation, can use saline water, works on any terrain, increases yield; best for orchards, vegetables, sugarcane, cotton. Disadvantages: high cost, clogging of emitters (filtration), rodent/ UV damage, salt build-up at edges.
- Micro-irrigation = drip + micro-sprinklers (PMKSY – "per drop more crop").

❖Comparison

Method	Efficiency	Best for
Border/ check	60–70%	Wheat, paddy, flat land
Furrow	60–75%	Row crops
Sprinkler	70–85%	Undulating land, sandy soil
Drip	90–95%	Orchards, vegetables, water-scarce

4. Quality of Irrigation Water

- Total dissolved salts (TDS)/ salinity: measured by electrical conductivity EC (dS/m or mmhos/cm).

Class	EC (µS/cm)	Suitability
C1 – Low	< 250	Excellent
C2 – Medium	250–750	Good
C3 – High	750–2250	Fair, salt-tolerant crops
C4 – Very high	> 2250	Poor

- **Sodium hazard – Sodium Adsorption Ratio (SAR):**

$$SAR = Na^{+} / \sqrt{[\,(Ca^{2+} + Mg^{2+})/2\,]}\ \text{(all in milli-equivalents/ litre)}$$

- SAR < 10 low hazard; 10–18 medium; 18–26 high; > 26 very high (soil dispersion, permeability ↓).
- **Percent sodium** (SSP) = (Na + K)×100/(Ca + Mg + Na + K); **> 60% harmful.**
- **Residual sodium carbonate** (RSC) = $(CO_3 + HCO_3) - (Ca + Mg)$: < 1.25 safe; > 2.5 unsuitable.
- **Toxic ions:** boron (> 2 mg/L toxic), chloride, fluoride; **heavy metals** in sewage water; pH 6.5–8.4 normal.
- **Suspended silt** (fertile but clogs), **pathogens/ organic load** (sewage – treated only), temperature.
- **Salt tolerance:** barley, cotton, sugar beet are tolerant; wheat, maize, rice are medium (semi-tolerant); beans, citrus and other fruit trees are sensitive.

5. Duty and Delta

Duty (D) = area of land (in hectares) that can be irrigated by a unit discharge of water (1 cumec = 1 m^3/s) running continuously throughout the **base period B** of the crop. Units: **ha/ cumec.**

Base period (B) = time (days) from first watering (preparatory) to last watering before harvest. **Delta (Δ)** = **total depth of water required by a crop during its base period** (depth if spread over the field) – in cm or m.

❖Relation between duty and delta

- 1 cumec for B days gives volume = 1 × B × 86400 m^3. This irrigates D hectares = D × 10^4 m^2:

$$\Delta = 8.64\,B / D\ (\Delta \text{ in metres; } B \text{ in days; } D \text{ in ha/cumec})\ [\Delta(cm) = 864\,B/D]$$

- **Derivation:** $\Delta = (B \times 86400\ m^3) / (D \times 10^4\ m^2) = 8.64\,B/D$ metres.

- Duty $\propto 1/\Delta$: higher duty means less water needed per hectare (better).

❖Duty at various places (not same everywhere because of losses)

- Duty at the field (at the field/ plot): discharge measured at the field – gives the highest duty value.
- Duty at the outlet / head of water course: discharge at the chak outlet – intermediate.
- Duty at the head of the canal (gross duty): discharge at canal head includes seepage & evaporation losses in channels, so for the same area the discharge is largest → duty is smallest.
- Hence $D_{field} > D_{outlet} > D_{head}$ (e.g., rice 800 ha/cumec at the field vs 600 at the canal head).

❖Example

> Wheat: $B = 120$ days, $\Delta = 40$ cm $= 0.4$ m → $D = 8.64 \times 120/0.4 = 2592$ ha/cumec. Rice: $B = 140$ days, $D = 900$ ha/cumec → $\Delta = 8.64 \times 140/900 = 1.34$ m.

> Canal discharge at outlet = 2 cumec, duty at outlet 2000 ha/cumec → area = 4000 ha.

❖Typical duties (ha/cumec at head)

Crop	Base period (d)	Duty
Rice	120–140	800–1000
Wheat	120	1800–2500
Sugar cane	320	700–900
Cotton	180	1400
Maize/ vegetables	100	1200

6. Factors Affecting Duty & Methods of Improving Duty

❖Factors affecting duty

- Type of crop & its base period; climate (temperature, humidity, rain, wind); soil type & permeability (sandy → low duty); method of irrigation (flood vs drip); canal conditions – unlined (seepage losses), length of canal, evaporation; characteristics of the canal system; skill of cultivators & water management; time & frequency of application; tillage/ ploughing; quality of water; topography & slope; water table; rotation of crops; delivery system (rotation, warabandi).

❖Methods of improving duty

- Lining the canals (reduces seepage, 30–40% loss saved); proper canal alignment & maintenance (de-silting, weed control); shorter canal length, better layout.
- Proper land levelling/ shaping, scientific irrigation methods (sprinkler/ drip); crop planning & rotation; conjunctive use of surface & ground water.
- Rotational water supply (warabandi), volumetric water charges, farmers' participation (Water Users' Associations – WUA), canal automation; deep ploughing & mulching; use of high-yielding, short-duration varieties; cut-off of canals during rain; ensure timely supply.

7. Water Requirement of Crops – Types of Soils

> Soil = the loose upper layer of earth crust formed by weathering of rocks + organic matter, supporting plant growth (consists of mineral particles, organic matter, water, air).

❖Types of soil (by texture – particle size)

- Gravel (> 2 mm), Sand (2 – 0.06 mm: coarse/ fine, high permeability, low water retention), Silt (0.06 – 0.002 mm), Clay (< 0.002 mm: high retention, low permeability, swells), Loam (mixture – ideal for agriculture: ~40% sand, 40% silt, 20% clay).
- Textural triangle classifies sandy loam, silty loam, clay loam etc.

❖Indian agricultural soils

Soil	Region	Features	Crops
Alluvial	Indo-Gangetic plains, deltas (Krishna, Godavari)	Fertile, loamy, rich in potash	Rice, wheat, sugarcane
Black (Regur/ cotton)	Deccan (Maharashtra, Telangana, A.P., Gujarat)	Clayey, retains moisture, swells/ cracks, rich in lime/ Fe, poor N,P	Cotton, sorghum, soybean
Red	Rayalaseema, Tamil Nadu, Odisha, Karnataka	Light, porous, acidic, poor N, P, lime	Millets, groundnut, pulses
Laterite	Western Ghats, Kerala, Assam	Leached, acidic, low fertility	Tea, coffee, cashew, rubber
Desert (arid)	Rajasthan	Sandy, saline	Bajra, millets (with irrigation)
Mountain/ forest	Himalayas	Humus-rich, acidic	Tea, fruits
Saline/ alkaline (Usar, Reh)	Punjab, UP, Haryana, canal commands	High salts, poor drainage	After reclamation

- A.P./ Telangana: red soils (≈ 50%), black soils (≈ 25%), alluvial deltas, sandy coastal soils.

8. Preparation of Land for Irrigation

- Clearing of trees, bushes, stones; land levelling & grading (laser levelling) to uniform slope for even flow; ploughing/ tilling to loosen soil, improve infiltration; land shaping – layout of channels (field channels, ditches), border strips, check basins, furrows; bunding (field boundaries); drainage provision; sowing with seed-bed preparation; fertiliser & manure application.
- Objectives: uniform distribution, minimum losses, high efficiency, prevent erosion & water-logging.

9. Soil Fertility & Soil–Water–Plant Relationship

❖Soil fertility

> Fertility = ability of soil to supply nutrients in adequate amount & balance for crop growth. Macro-nutrients: N, P, K (primary), Ca, Mg, S; micro-nutrients: Fe, Zn, Mn, Cu, B, Mo, Cl.

- Maintained by **manures, fertilisers, crop rotation (legumes fix N), green manure, proper irrigation; soil testing, pH 6.5–7.5 ideal; fertigation.**

❖Soil–water–plant relationship

- **Soil-water phases:** solids (45%), water (25%), air (25%), organic (5%) – ideal loam.
- Water in soil is held against gravity by **adhesion & cohesion** (surface tension) in **films & capillary pores;** plant roots absorb **capillary water (available water).**
- **Types of soil water:**
  - **Gravitational water** – drains freely by gravity (after irrigation/ rain); not available (lost in ~ 2–3 days).
  - **Capillary water** – held in pores by surface tension; **available to plants.**
  - **Hygroscopic water** – thin film on soil particles by adhesion; **unavailable.**
  - **Chemically combined water** (in minerals).
- **Moisture content** (dry-weight basis) $w = (W_w / W_s) \times 100$; volume basis $\theta = V_w/V$.

10. Vertical Distribution of Soil Moisture

- After irrigation/ rain: **zones from top: soil–water (root) zone → intermediate zone → capillary fringe → saturated zone.**
- Moisture is **highest at the surface just after wetting,** decreases with depth; **plants extract mostly from the upper root zone (≈ 40% from 1st quarter, 30%, 20%, 10% from the lower quarters of root depth – "40-30-20-10 rule").**
- Root depth: shallow (rice, vegetables 30–60 cm), medium (wheat, maize 90–120 cm), deep (cotton, sugarcane, 1.5–2 m).

11. Soil Moisture Tension & Stress

- **Soil moisture tension (suction) = force (negative pressure) with which water is held by the soil;** expressed as **atmosphere, bar, cm of water, kPa, or pF** ($pF = \log_{10}$ of tension in cm of water).

> Tension ↑ as soil dries; saturation ≈ 0 ; field capacity ≈ 1/3 atm (pF 2.5) ; PWP ≈ 15 atm (pF 4.2)

- Measured by **tensiometer** (porous cup, vacuum gauge, range 0–0.8 atm), gypsum blocks, neutron probe, pressure-plate apparatus.

- Soil moisture stress = combined effect of tension & osmotic (salt) forces that reduces availability of water to plants; plant wilts when stress exceeds root suction.
- Total stress (potential) = matric (tension) + osmotic + gravitational components.

12. Soil Moisture Constants

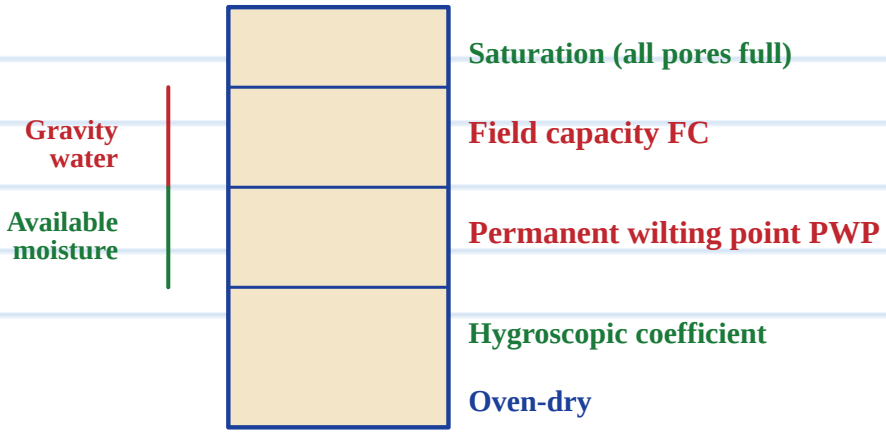


Fig 3.1 Soil moisture constants

- Saturation capacity: all pores filled; maximum water content (moisture tension ≈ 0).
- Field capacity (FC): water held by soil after free (gravitational) drainage has ceased (≈ 2–3 days after irrigation); tension ≈ 1/3 atm; upper limit of available water.
- Permanent wilting point (PWP / wilting coefficient): moisture content at which plants wilt permanently & cannot recover (tension 15 atm); lower limit of available water.
- Available moisture (AM) = FC – PWP (readily available portion ≈ 50–75% of it).
- Hygroscopic coefficient: moisture absorbed from saturated air, held at tension ≈ 31 atm; ultimate wilting point ≈ hygroscopic.
- Moisture equivalent, optimum moisture, readily available moisture.

❖Typical values (% of dry weight)

Soil	FC	PWP	Available
Sand	10	4	6
Sandy loam	18	8	10
Loam	27	13	14
Clay loam	35	18	17
Clay	40	24	16

❖Limiting soil-moisture conditions

- Upper limit = field capacity (beyond this water is wasted by percolation).
- Lower limit = permanent wilting point; in practice irrigation is given when 50–60% of available moisture is used – "optimum moisture/ allowable depletion (p)" (before reaching PWP; crop yield falls beyond "critical moisture").
- Readily available moisture = 0.5–0.75 × (FC – PWP).
- Over-irrigation → aeration problem (waterlogging); under-irrigation → stress.

13. Depth & Frequency of Irrigation

Depth of water per irrigation $d = [(F_c - PWP)/100] \times (\gamma_d/\gamma_w) \times D \times p$
(or $d = (F_c - M_0)/100 \times \gamma_d/\gamma_w \times D$)

- F_c = field capacity (%), PWP (%), γ_d = dry unit weight (specific gravity) of soil, γ_w = unit weight of water, D = root-zone depth, p = allowable depletion fraction (0.5–0.6), M_0 = moisture at which irrigation is given.

Frequency (interval) of irrigation $f = d / C_u$ (days), C_u = peak consumptive use (mm/day)

❖Example

Root zone D = 80 cm, F_c = 28%, PWP = 14%, bulk density ratio γ_d/γ_w = 1.4, p = 0.5 → $d = 0.14 \times 1.4 \times 80 \times 0.5 = 7.84$ cm. Peak use = 0.5 cm/day → f = 7.84/0.5 ≈ 15 days.

14. Command Area Terms

- Gross Command Area (GCA): total area enclosed by the boundaries that can be irrigated by a canal system, whether or not cultivable (includes roads, villages, ponds, barren land).
- Culturable Command Area (CCA): the part of GCA that is fit for cultivation (excludes unfit land, habitation). CCA = GCA – unculturable area.
- Culturable cultivated area: CCA actually cultivated at a time (sown). Culturable uncultivated area: CCA not cultivated (fallow/ waste, but fit). Intensity of irrigation = irrigated area/ CCA × 100%.

GCA = CCA + Unculturable area ; CCA = Cultivated + Uncultivated (culturable)

- Irrigation capacity of canal designed on CCA × intensity; outlet serves a chak.

15. Kor Depth and Kor Period

> Kor watering = the first watering (usually applied to a crop when it is a few cm tall) – generally the heaviest and given to soften soil & for germination.

- Kor depth = depth of water applied for the first watering (e.g., wheat ≈ 13.5 cm, rice ≈ 19 cm, sugar cane ≈ 16.5 cm, cotton ≈ 10 cm).
- Kor period = time (days) for which the first (kor) watering is applied/ the period during which kor watering is given in the whole canal command (wheat ≈ 4 weeks; rice ≈ 2.5 weeks) – the canal is designed on the peak kor requirement.

Outlet discharge $Q = A \times d / (B_{kor}) \approx (A \times \text{Kor depth}) / (\text{kor period} \times 86400)$

❖Example

> Wheat area A = 2000 ha, kor depth = 13.5 cm, kor period 4 weeks (28 days): Volume = $2000 \times 10^4 \times 0.135 = 2.7 \times 10^6\ m^3$; $Q = 2.7 \times 10^6 / (28 \times 86400) = 1.12\ m^3/s$.

16. Crop Seasons & Crop Rotation

❖Crop seasons in India

Season	Sowing – harvest	Crops
Kharif (monsoon)	June–July → Sept–Oct	Rice (paddy), maize, jowar, bajra, cotton, groundnut, sugarcane (annual)
Rabi (winter)	Oct–Nov → Mar–Apr	Wheat, barley, gram, peas, mustard, sunflower
Zaid (summer)	Mar → June	Watermelon, cucumber, vegetables, fodder

- Perennial/ two-seasonal crops: sugarcane (12–18 months), banana, cotton (8 months).
- Different crops have different water requirement (paddy 1200–2500 mm; wheat 450–650; sugarcane 1500–2500; cotton 700–1300; maize 500–800; pulses 250–500 mm).

❖Crop rotation

> Crop rotation = growing different crops in succession on the same land in a planned sequence.

- **Advantages:** maintains soil fertility (legumes fix nitrogen), controls pests, weeds and diseases, balances nutrient removal & root depth (shallow/ deep rooted), improves soil structure, reduces salinity, reduces water requirement variation, better yields, risk spreading.
- **Examples:** rice–wheat, rice–pulses, cotton–groundnut, paddy–groundnut–green gram, wheat–fallow–sugarcane. Fallowing, mixed & inter-cropping.

17. Irrigation Efficiencies

> Efficiency = ratio of output water to input water at various stages (water is lost by seepage, evaporation, deep percolation, runoff).

Conveyance (canal) efficiency $\eta_c = W_f / W_r \times 100$ (water delivered to the field / water diverted at river head)

Water application efficiency $\eta_a = W_s / W_f \times 100$ (water stored in root zone / water delivered to the field)

Water storage efficiency $\eta_s = W_s / W_n \times 100$ (stored / water needed to bring root zone to FC)

Water use efficiency $\eta_u = W_u / W_d \times 100$ (consumed by crop (consumptive use)/ water delivered)

- Project (overall) efficiency = $\eta_c \times \eta_a$ (≈ 40–60% for flood systems; drip 90%).
- Water distribution efficiency $\eta_d = [1 - (y/d)] \times 100$ (y = average numerical deviation from mean depth d).
- Typical: η_c 60–80%; η_a 50–70% (surface), 75% sprinkler, 90% drip.

❖Example

> Water diverted at head = 5 m^3/s, delivered to field = 3.5 m^3/s → η_c = 70%.
> Water stored in root zone = 2.8 → η_a = 2.8/3.5 = 80%; overall = 0.7×0.8 = 56%.

18. Determination of Irrigation Requirements of Crops

❖Terms

- **Consumptive use (C_u)/ Evapotranspiration:** water used by crop in transpiration + building plant tissue + evaporation from soil/ leaf surface.

- Effective rainfall (R_e): portion of rain stored in root zone & useful (≈ 70–80% of rain above 5–6 mm; 0 if too little/ too much).
- Net irrigation requirement (NIR) = C_u – R_e – (soil moisture & ground-water contribution) + leaching requirement (if saline).

> NIR = C_u – R_e – S; Field irrigation requirement FIR = NIR / η_a; Gross irrigation requirement GIR = NIR / (η_a η_c)

- Add water for land preparation (puddling – paddy: 150–250 mm), nursery, percolation losses (paddy: 2–6 mm/day), leaching.

❖Assessment of irrigation water

- Water is charged to cultivators by:
  - Volumetric (by volume) assessment – metering at outlet (fairest; modules like Gibb's, Kennedy's).
  - Area (acreage) assessment – crop-wise rate per hectare (most common in India; rates depend on crop & season, e.g., paddy higher than dry crop).
  - Seasonal/ per-irrigation rates, permanent assessment (fixed per ha irrespective of use, whether or not used), Batai (share of crop), block system, Warabandi (fixed rotation turn schedule by time).

❖Warabandi (rotational water supply) & water charges

- Fixed day/ hour schedule for each farmer – equitable distribution, reduces conflict & wastage; used in Punjab/ Haryana; A.P. "Neeru-Pragathi" water management under WUAs.

19. Consumptive Use of Water

> Consumptive use (C_u) = total water (depth) needed by a crop from sowing to harvest for transpiration + evaporation (evapotranspiration, ET). Expressed as depth (mm/ day, mm/ season) and used to find delta.

❖Factors affecting consumptive use

- Climate: temperature, humidity, wind, sunshine hours/ solar radiation, rainfall.
- Crop: type, growth stage (peak at flowering), crop period, leaf area, density, root depth.
- Soil: moisture availability, soil texture & water table depth, fertility, salinity.
- Cultural/ management: irrigation method & frequency, tillage, mulch, planting date.

- Latitude, season, altitude.

❖Direct measurement

- Tank (lysimeter) experiments: crop grown in large tank/ drum with known water inputs; ET = water added – drained (lysimeter weighing).
- Field plot experiments: measure water applied + rainfall + change in soil moisture – drainage.
- Soil-moisture studies (gravimetric/ neutron probe) in the root zone; integrating method (water balance in command area).
- Inflow–outflow studies for large areas.

❖Determination by equations (theory)

- (1) Blaney–Criddle formula (USBR):

$$C_u = \Sigma\, k\, f,\ f = p\,(0.46\,T + 8.13)\ \text{(mm/day for daily p)}$$

  - T = mean temperature (°C); p = % of annual daytime hours occurring in the period (≈ 0.27 per day, 8–9 per month); f = consumptive-use factor; k = crop coefficient (e.g., 0.8–1.2 for different crops/ season).
- (2) Penman equation (energy balance + aerodynamic) – $ET_0 = [\Delta\,(R_n - G) + \gamma \cdot 6.43\,(1 + 0.536\,u)\,(e_s - e_a)\,] / [\lambda\,(\Delta + \gamma)\,]$; FAO Penman–Monteith (standard): ET_0 reference crop; $ET_c = K_c \times ET_0$.
- (3) Hargreaves: $ET_0 = 0.0023\,R_a\,(T + 17.8)\,(T_{max} - T_{min})^{0.5}$ (needs only temperature).
- (4) Thornthwaite, Christiansen, pan-evaporation method: $ET_0 = K_p \times E_{pan}$ ($K_p \approx 0.7$–0.8).
- (5) Lowry–Johnson, Hargreaves–Samani, Jensen–Haise.

❖Example

Blaney–Criddle (daily form, p = daily % of annual daytime hours ≈ 0.27): T = 28 °C → f = 0.27 (0.46×28 + 8.13) = 0.27 × 21.0 = 5.67 mm/day; k = 0.9 → C_u = k f = 5.1 mm/day (≈ 612 mm for a 120-day crop).

Simple calculation: crop period 120 days, ET_0 average 5 mm/d, K_c = 1.0 → C_u = 600 mm; effective rain 150 mm → NIR = 450 mm; with η_a = 60% & η_c = 75% → GIR = 450/(0.6×0.75) = 1000 mm.

20. Solved Problems

❖Problem 1 – Delta

> $B = 110$ days, $D = 1600$ ha/cumec → $\Delta = 8.64 \times 110/1600 = 0.594$ m ≈ 59.4 cm.

❖Problem 2 – Canal discharge for CCA

> CCA = 10,000 ha under wheat (delta 45 cm, base 120 d): $D = 8.64 \times 120/0.45 = 2304$ ha/cumec → $Q = 10000/2304 = 4.34$ m^3/s (at outlet).

❖Problem 3 – Efficiency-based requirement

> Net field requirement = 6 cm over 100 ha = 6000 m^3; $\eta_a = 60\%$ → delivered = 10,000 m^3; $\eta_c = 75\%$ → at head = 13,333 m^3.

❖Problem 4 – Frequency

> $d = 6$ cm, peak $C_u = 0.6$ cm/day → $f = 10$ days.

21. Important Questions – Unit III

- Necessity, importance, advantages & ill effects of irrigation. Types of irrigation.
- Describe methods of application: border, check basin, furrow, sprinkler, drip.
- Quality of irrigation water (SAR, EC); define duty & delta and derive $\Delta = 8.64B/D$; factors affecting duty & methods of improving.
- Soil moisture constants; depth & frequency of irrigation; GCA, CCA, kor depth/ period.
- Crop seasons & rotation; irrigation efficiencies; consumptive use & equations; assessment of irrigation water.

$$\Delta = 8.64B/D \; ; \; d = (Fc - PWP)(\gamma d/\gamma w)D \; ; \; f = d/Cu \; ; \; GIR = NIR/(\eta a \, \eta c)$$